

CSE446: Blockchain & Cryptocurrencies

Lecture – 18: Blockchain Properties, Misconceptions, Limitations, Feasibility

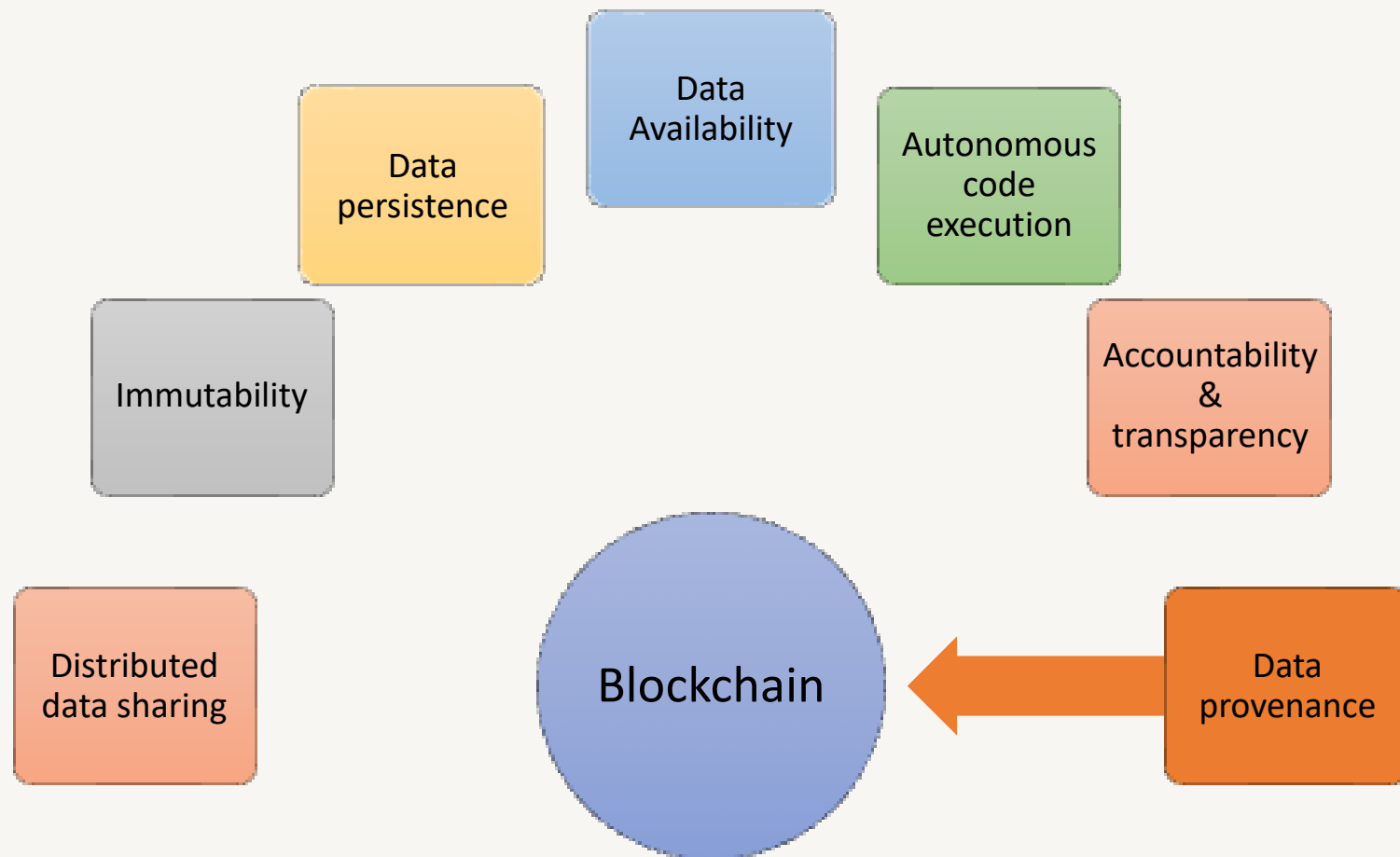


Inspiring Excellence

Agenda

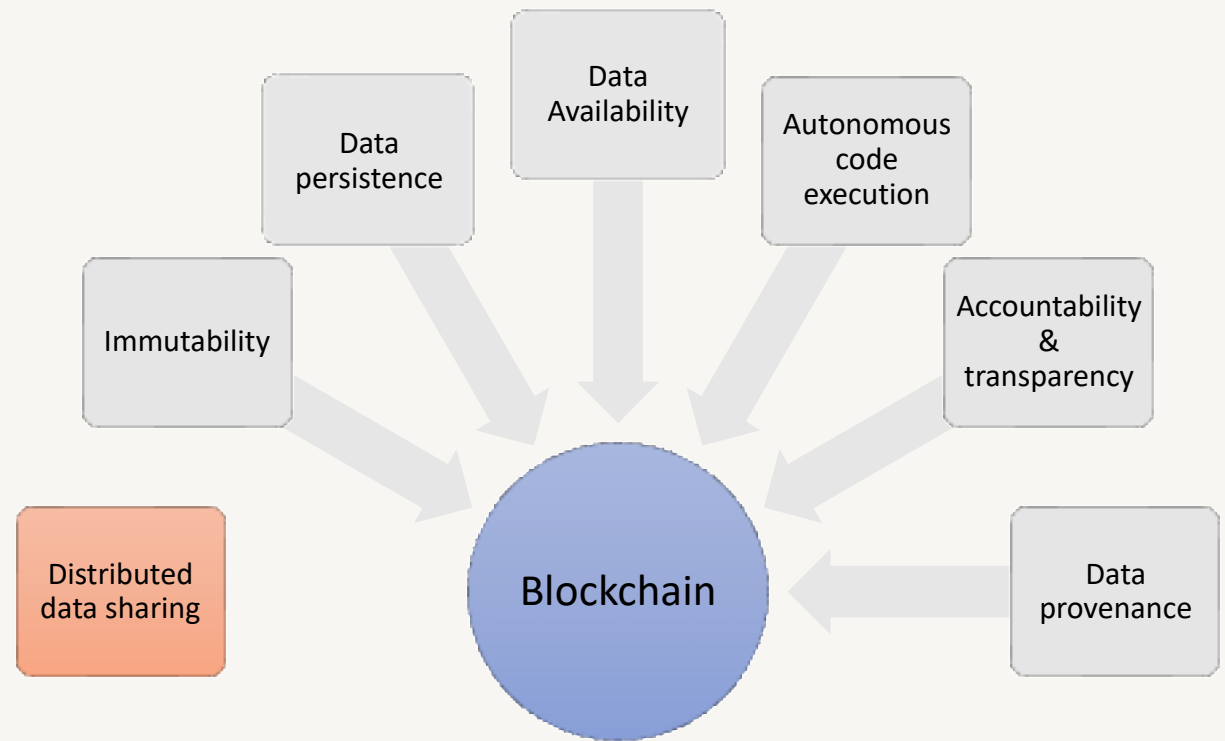
- Blockchain Properties
- Blockchain Misconceptions
- Blockchain Limitations
- Blockchain feasibility

Blockchain properties



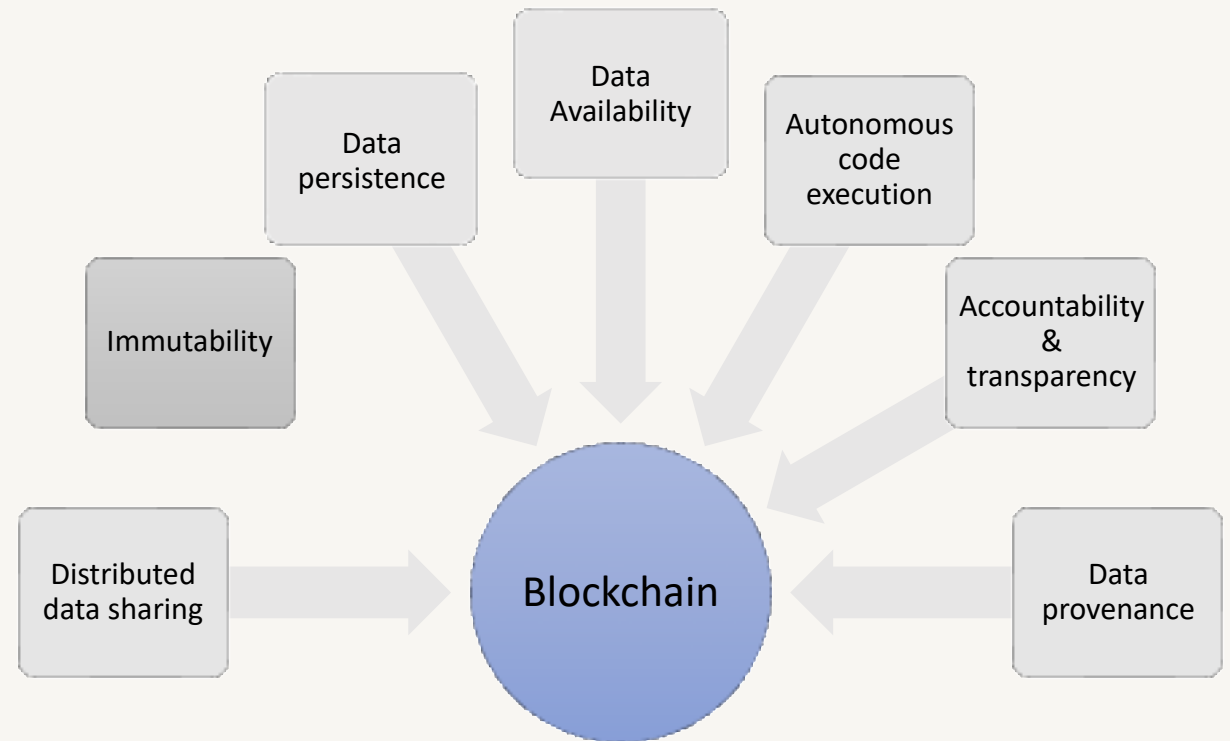
Distributed Data Sharing

- Blockchain data is distributed across multiple nodes
- The protocol ensures that data inserted in a particular node gets synced across all nodes in a timely fashion



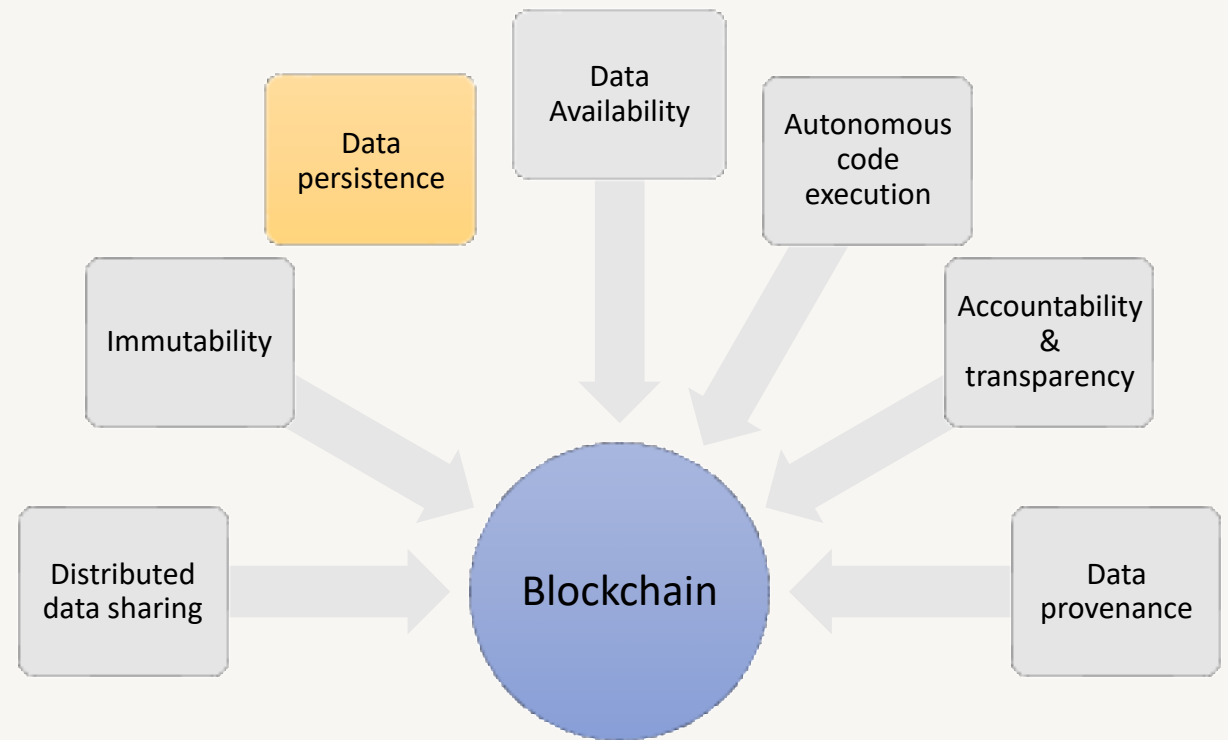
Immutability

- Data and code immutability in blockchain emerges from the fact that to change data/code inserted in a previous block, an attacker must possess either
 - significant computational power, in case of a public blockchain
 - or compromise the majority of nodes in any type of blockchain



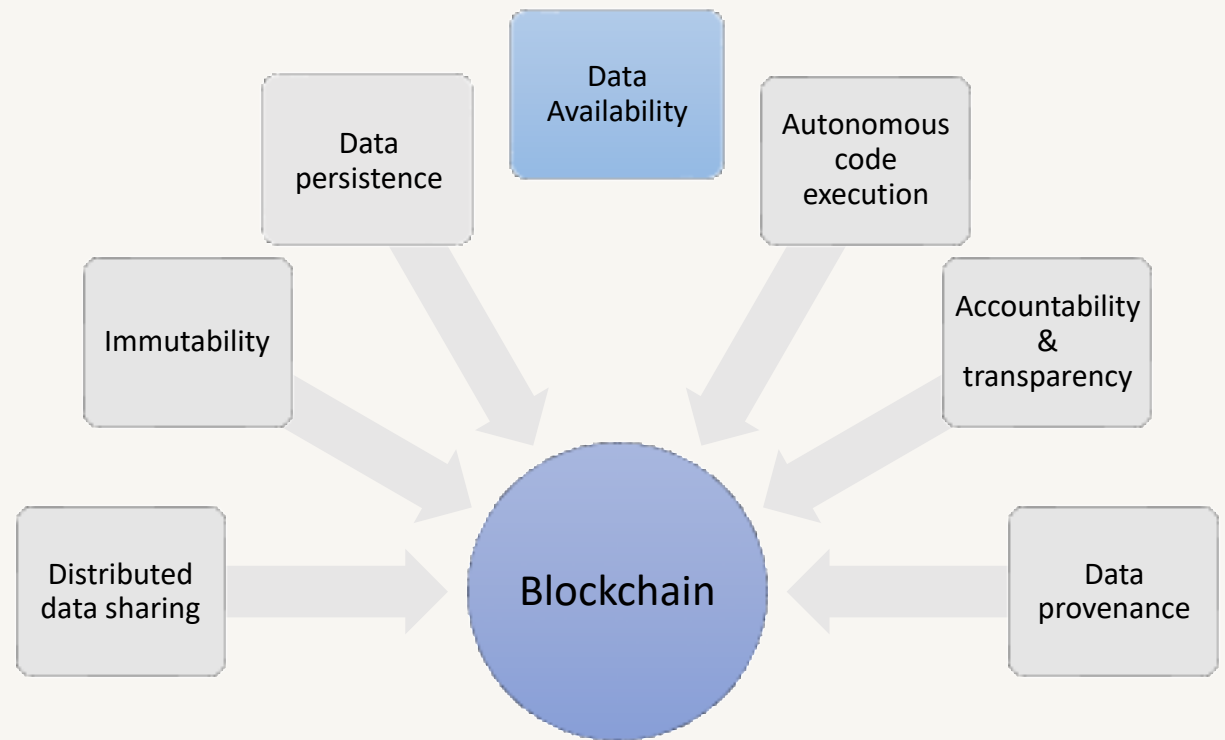
Data persistence

- Being a distributed system implies that data in a blockchain will persist as long as there are enough nodes to execute the protocol in a secure way



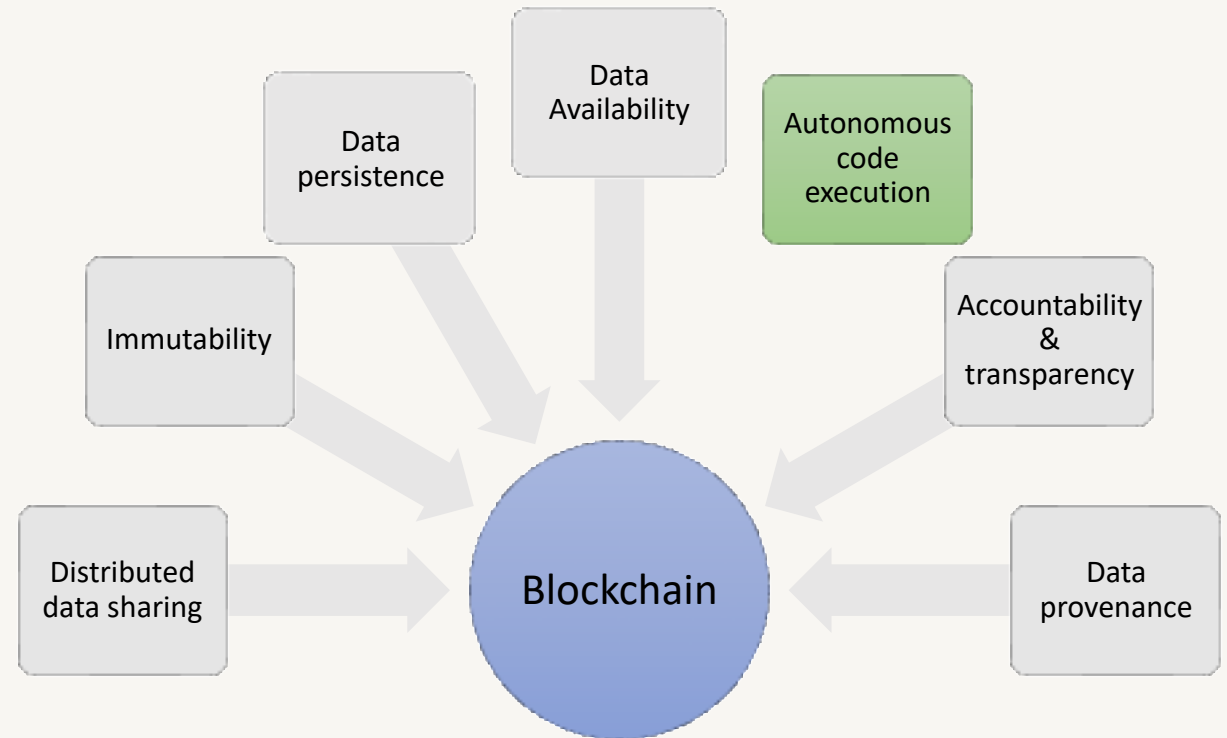
Data availability

- Data in blockchain are always available
- Even when a particular node is offline, data can be retrieved from another node in the blockchain



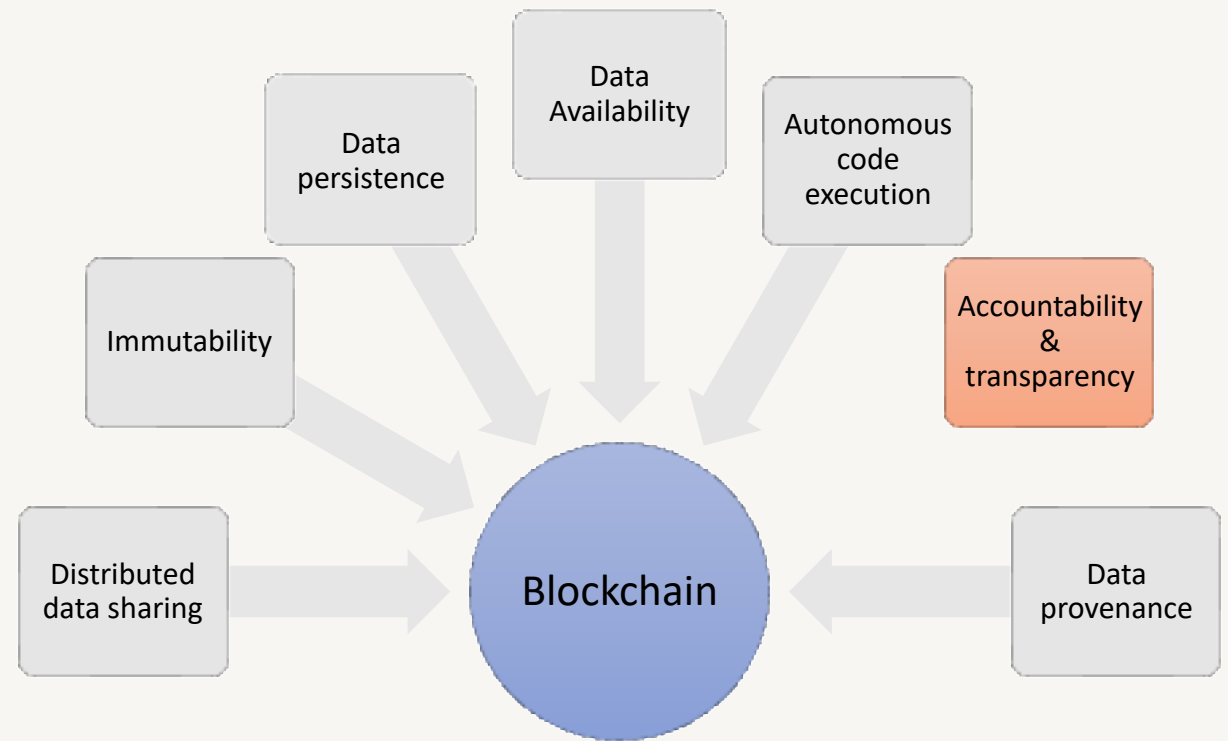
Autonomous code execution

- A smart-contract will facilitate autonomous code execution without a single point of failure
- It does not require any human intervention
 - anyone can submit a transaction to execute a code
- For any private blockchain system, anyone authorised can execute a code



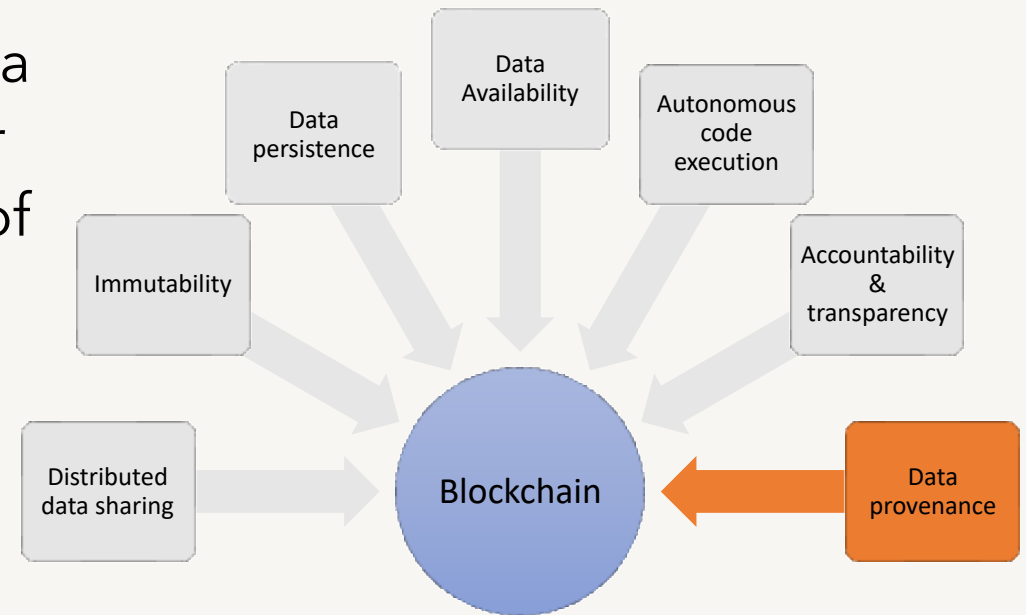
Accountability and transparency

- All authorised entities can verify each single transaction which can ensure accountability and transparency



Data provenance

- The term “data provenance” refers to a record trail that accounts for the origin of a piece of data (in a database, document or repository) together with an explanation of how and why it got to the present place
- Data in a blockchain can only be stored with a signed transaction
- Blockchain also stores the transactions which might have changed the data
- Both of these ensure data provenance



Blockchain misconception



DATA
IMMUTABILITY



LARGE-SCALE
DATA STORAGE



DATA INTEGRITY



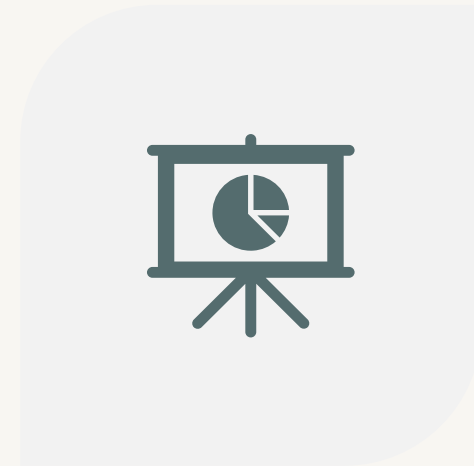
DATA
ENCRYPTION



POWER
CONSUMPTION

Data immutability misconception

- Blockchain data can be never be changed
- This is true for transaction/blockchain data which are immutable
- However, smart-contract data can be changed as required
 - Remember we could change different variable values in a smart-contract
 - However, how such data is changed is recorded in the blockchain and hence, is immutable



DATA IMMUTABILITY

Large-scale data storage misconception

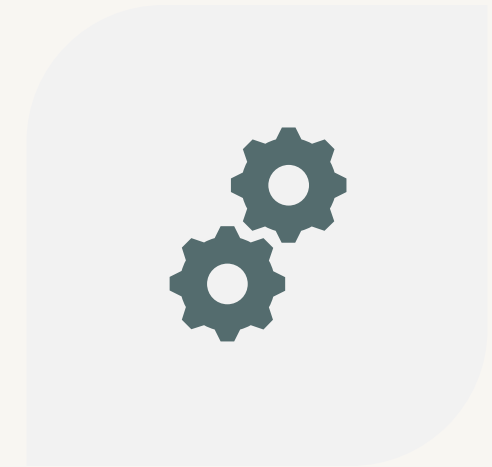
- Blockchain provides integrity of data and hence, users are tempted to store large amount of data in blockchain to ensure integrity
- Performance of any database in terms of data access rate is much better than that of any blockchain system
- Also storing a large amount of data in a public blockchain is costly
- Thus, it is advisable to store as minimum data as possible in the blockchain



LARGE-SCALE
DATA STORAGE

Data integrity misconception

- People think blockchain can support data integrity for any type of data
- However, it must be remembered that a blockchain system is essentially a "*Garbage-in-garbage-out*" system
- A corrupted data will be stored and remain as corrupted
- It can guarantee the integrity of data only after it is stored in the blockchain



DATA INTEGRITY

Data encryption misconception

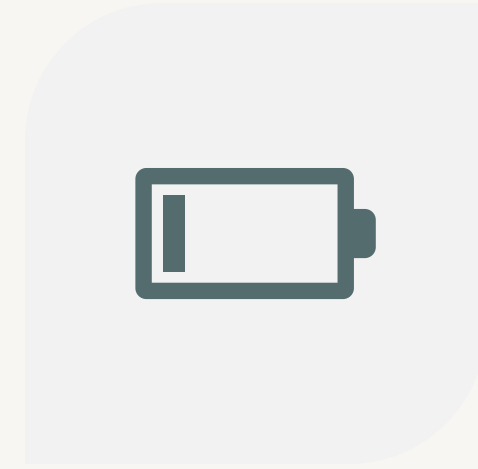
- Many believe that a blockchain provides data encryption by default
- A blockchain system strongly depends on cryptographic mechanisms, such as digital signature and cryptographic hash, to function
- Digital signature is used for data provenance while a cryptographic hash is used to ensure data integrity
- In a blockchain system, data encryption is not provided



DATA
ENCRYPTION

Power consumption misconception

- Many believe that every blockchain system consumes a huge amount of power
- However, the reality is that only public blockchain systems which utilise PoW or similar consensus algorithms consume huge electricity
- Public blockchain systems with PoS or DPoS consume significantly less electricity
- The power consumption of any private blockchain system will be comparable to any existing system



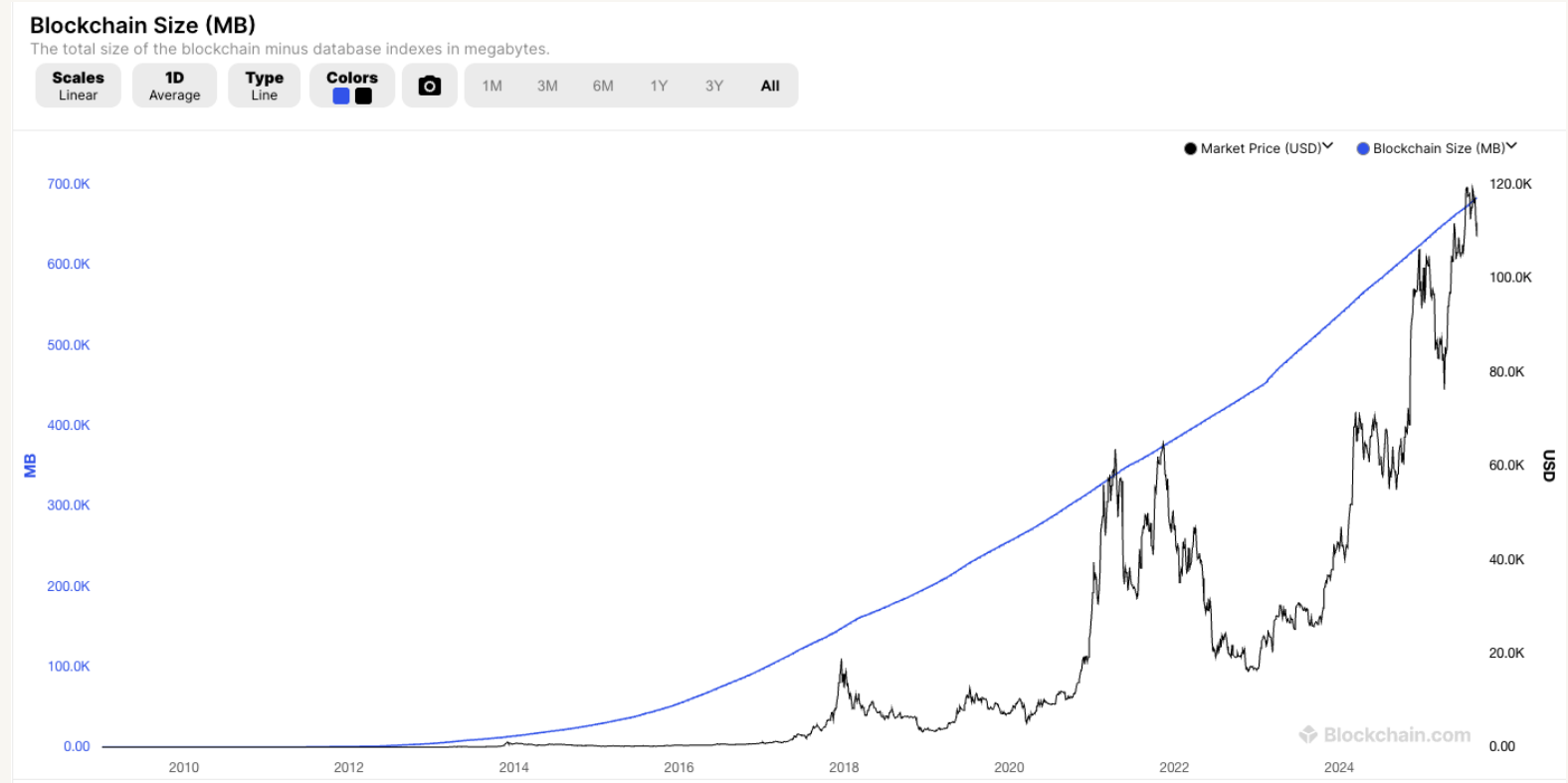
POWER
CONSUMPTION

Blockchain limitations

- Blockchain bloating
- Blockchain scaling
- Security vs Decentralisation vs Scaling
- Power consumption (already covered)
- Code immutability
- Usability
- Associated expense

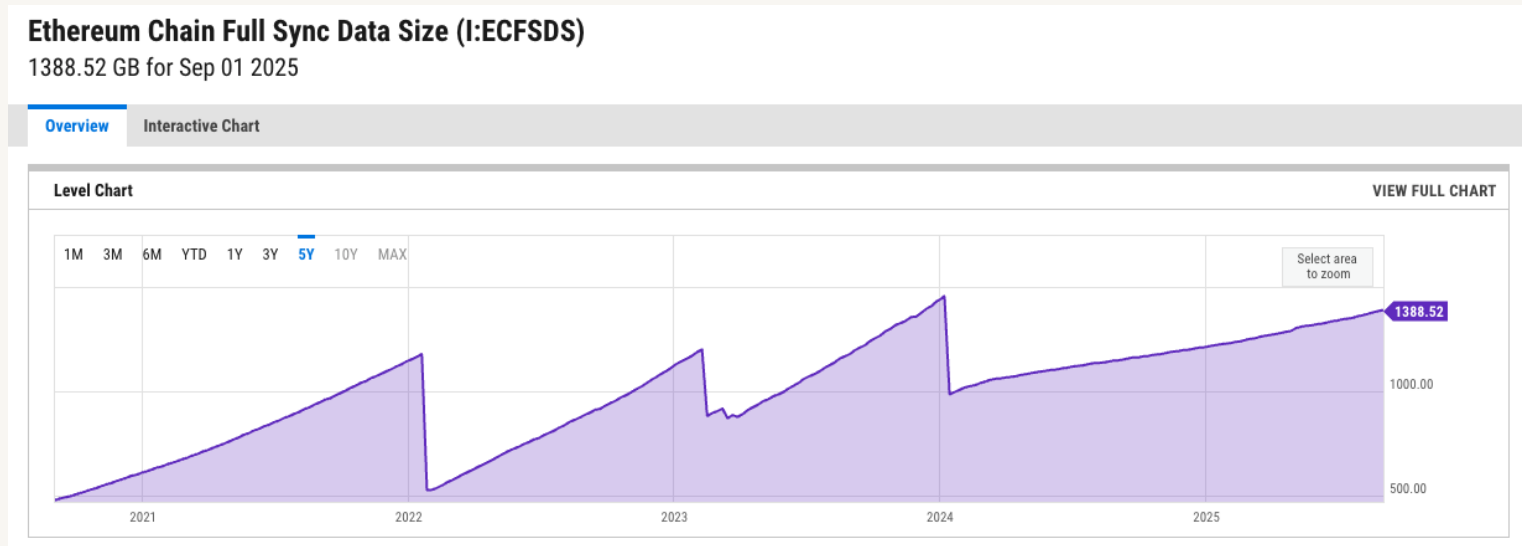
Blockchain bloating

- Blockchain being an add-only distributed database, its size keeps increasing
- Bitcoin size is currently nearly 700 GB and increasing



Blockchain bloating

- Ethereum size is currently >1TB and increasing
- What will happen in 20/30/50 years time?
- Blockchain bloating would also increase data processing time
 - Finding a particular UTXO and so on

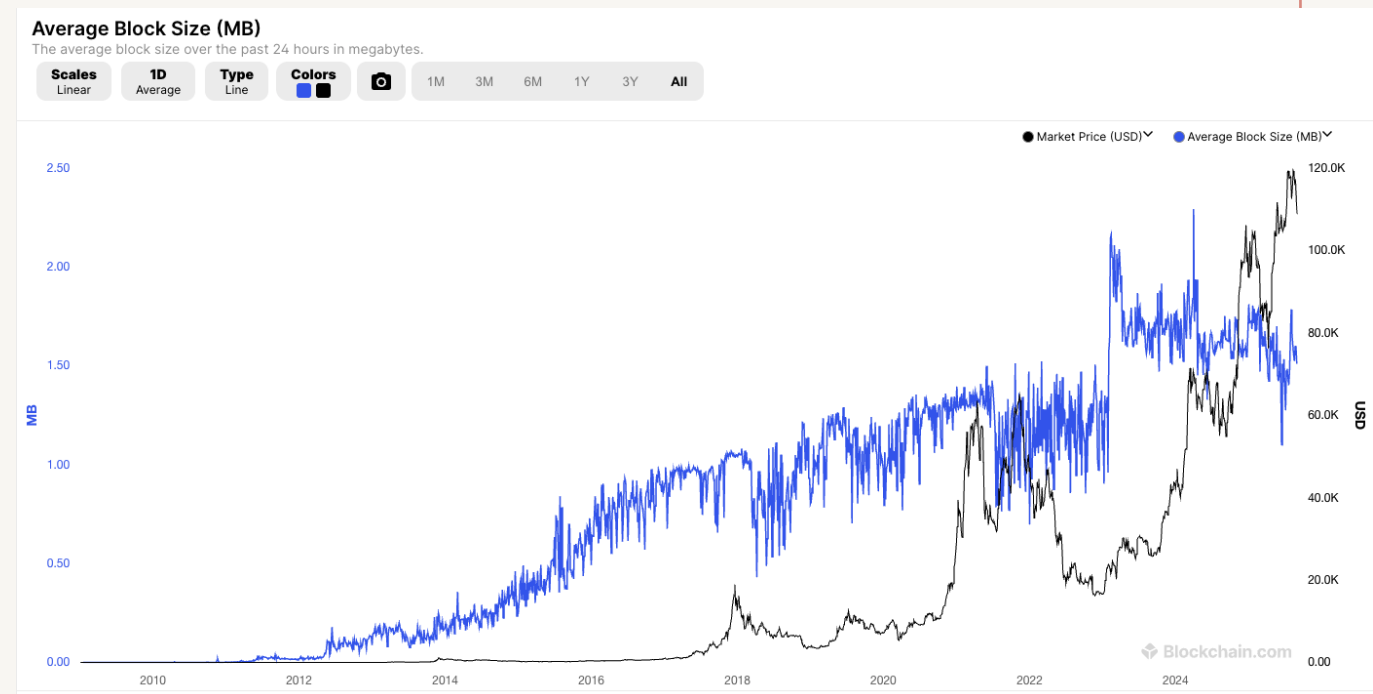


Blockchain scaling

- The blockchain scalability problem refers to the limited capability of the blockchain network to handle large amounts of transactions on its platform in a short span of time
- This limited capability is due to two reasons:
 - Limitations in block size
 - Limited TPS (transaction per second)

Blockchain scaling

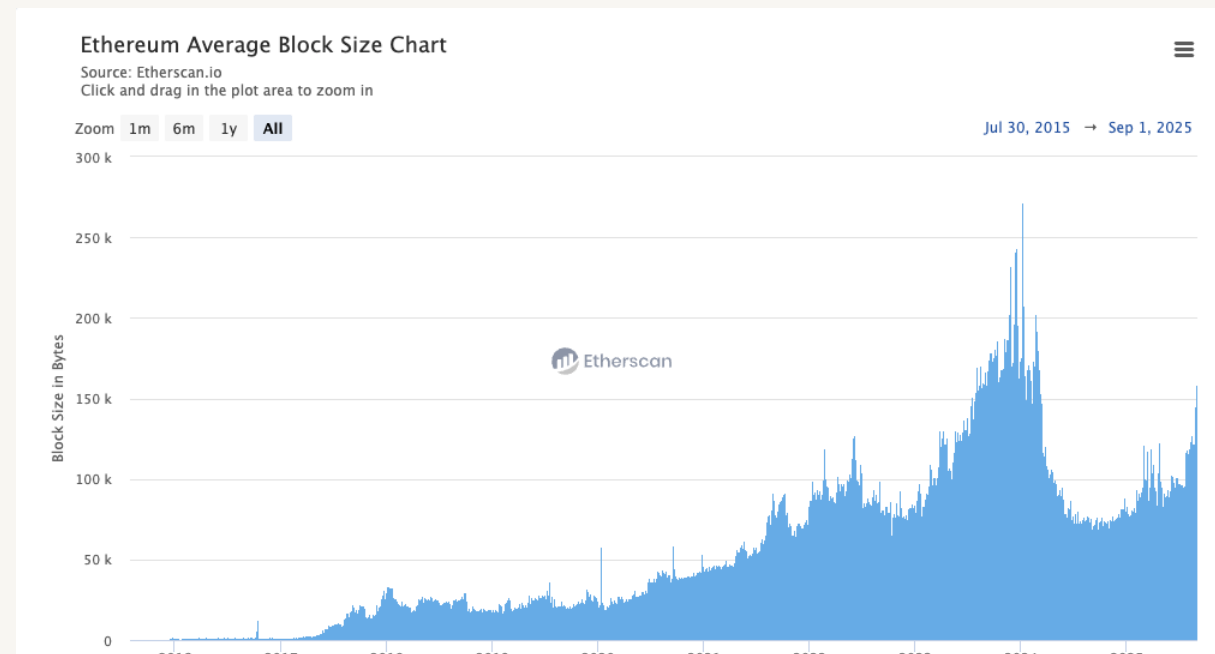
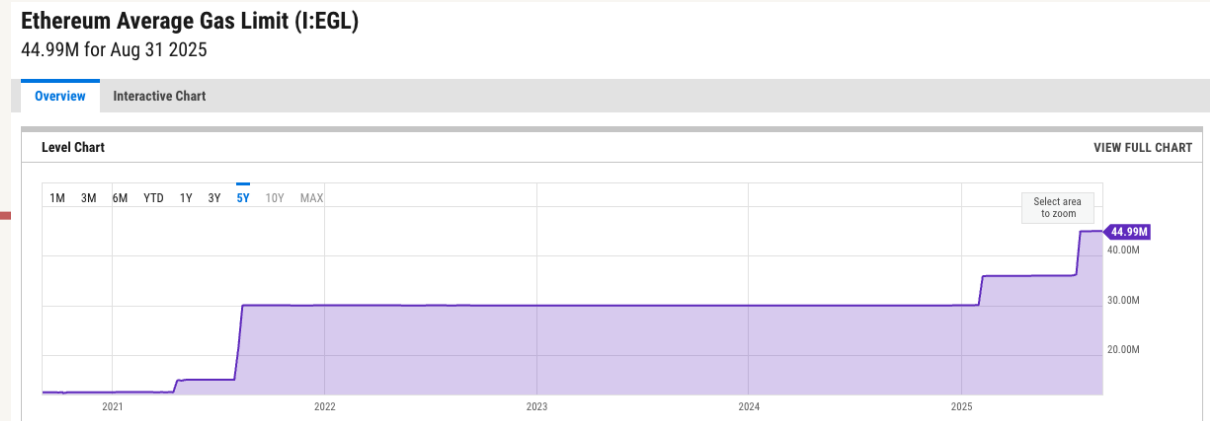
- Bitcoin had a limitation of 1MB block size, currently around 2MB
- That means the transactions need to be selected in such a way so that total block size including header is around 2 MB
- Also, the TPS in Bitcoin is around 3-7
- Both of these imply that Bitcoin finds it difficult to handle a large number of transactions in a short period of time



Blockchain scaling

- In Ethereum, each block has a target size of around 45 million gas
- The size of blocks will increase or decrease in accordance with network demands
- Currently, it results in 1.5MB block size in average
- TPS in Ethereum is around 18

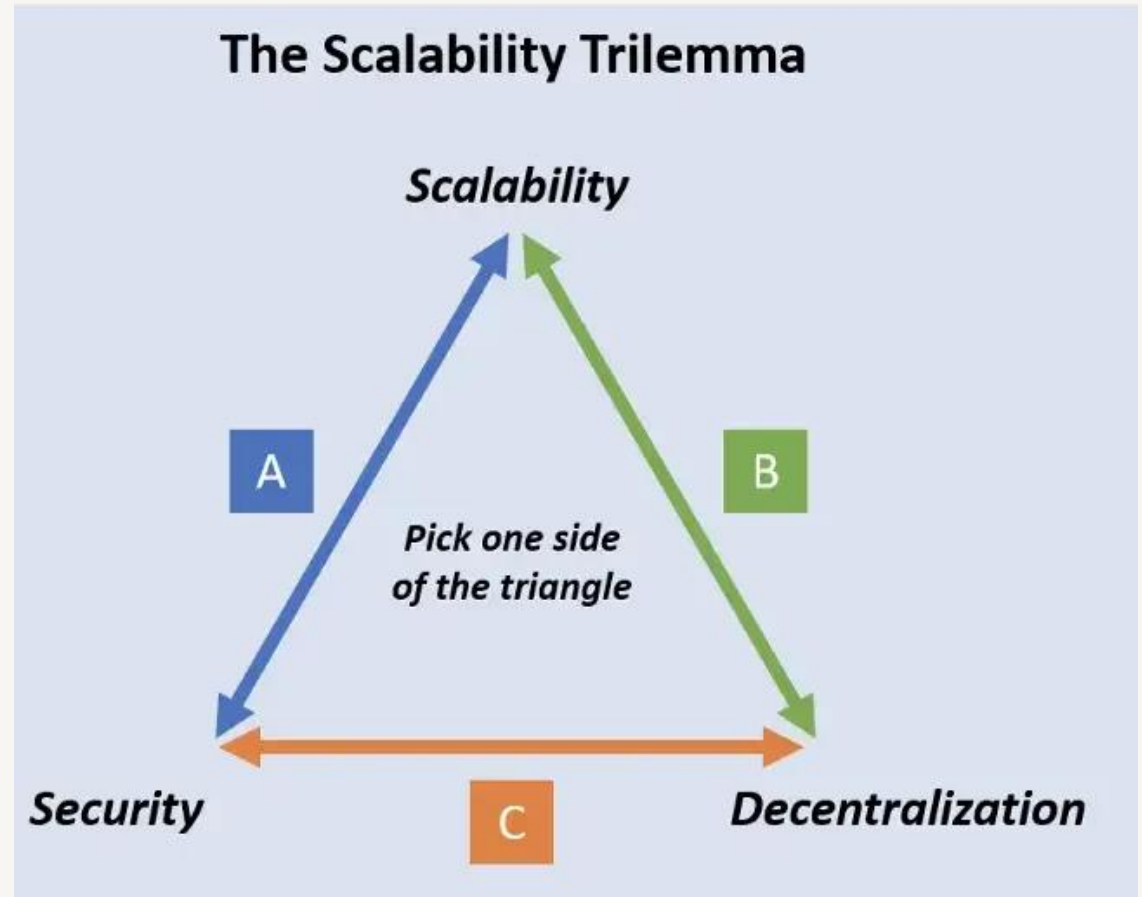
https://ycharts.com/indicators/ethereum_average_gas_limit



<https://etherscan.io/chart/blocksize>

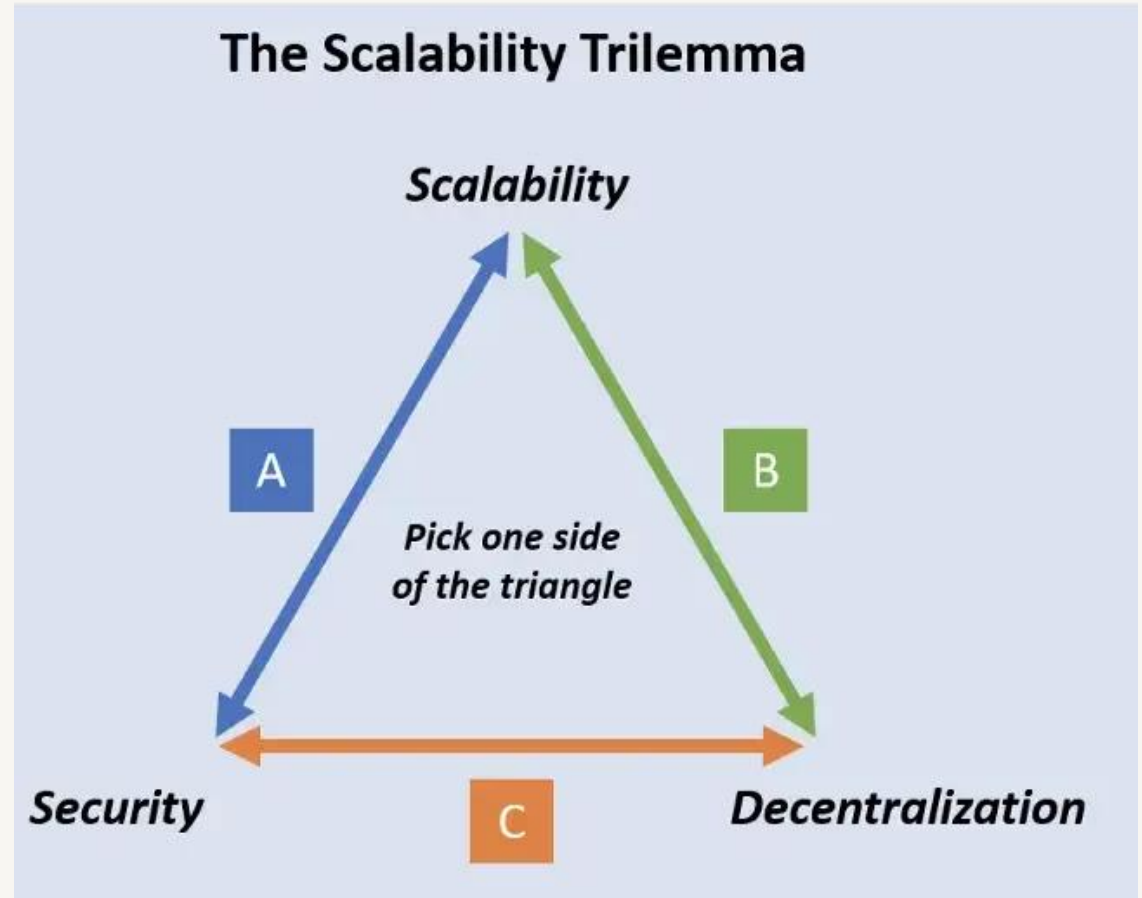
Security vs Decentralisation vs Scaling

- Blockchain scalability trilemma was introduced by Vitalik
 - "trilemma" is a situation where you can only get two out of three desirable outcomes
- The trilemma is we can only choose any of the two properties from these three
 - Security and Scalability, **Security and Decentralisation**, Decentralisation and Scalability



Security vs Decentralisation vs Scaling

- Decentralisation implies not utilising a single central system
- This will also increase the security
 - It is not a single point of failure anymore
- However, this might reduce the TPS
 - As it would require slightly more time to reach a consensus
 - Remember, more people mean more arguments and difficult to reach a consensus



Code immutability

- Once a smart-contract is deployed it becomes immutable
- This has huge advantage; however, it also introduces limitations
- If there is a bug in a smart-contract it cannot be rectified
- The error needs to be fixed off-chain and then re-deployed in the network
 - This will result in a new contract address
- The Dapp then needs to be updated with the new contract address

Usability

- Remember using Metamask setup?
- Need to remember the password or safely store the mnemonic words (passphrase)
 - If you forget your password and don't safely backup your passphrase then you cannot recover your wallet or funds in it
- Think about, how did you find it: easy or difficult?
- Then think about the general people and how they would find it to use such a complex process?
- Researchers have found strong usability issues with Blockchain wallets

Associated expense

- It takes considerable investment to join the mining and staking process
- Storage and computation costs crypto-currency (eth)
- In a 2018 estimation:
 - When 1 ETH = 200 USD
 - 1 KB required 2 USD
 - 1 MB would cost around 2000 USD
- But now, 1 ETH = 3000 USD
 - It is very difficult to predict, how much it might cost
 - One option is to try via Ganache or in the test network

Blockchain feasibility



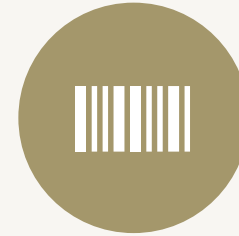
DECENTRALI
SATION



DISINTERME
DIATION



P2P VALUE
TRANSFER

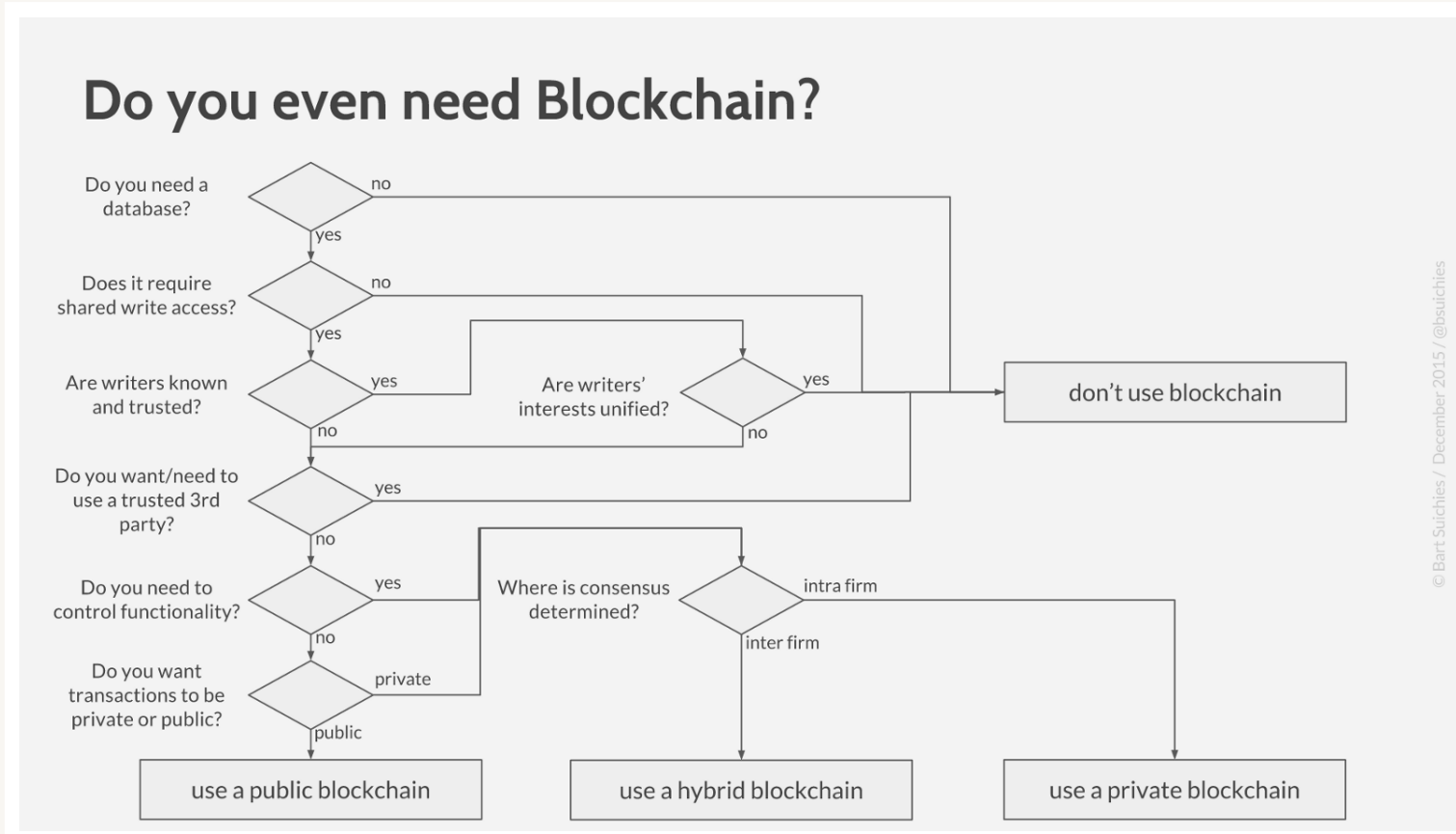


DATA/CODE
IMMUTABILITY



DISTRIBUTED
DATA SHARING

Blockchain feasibility

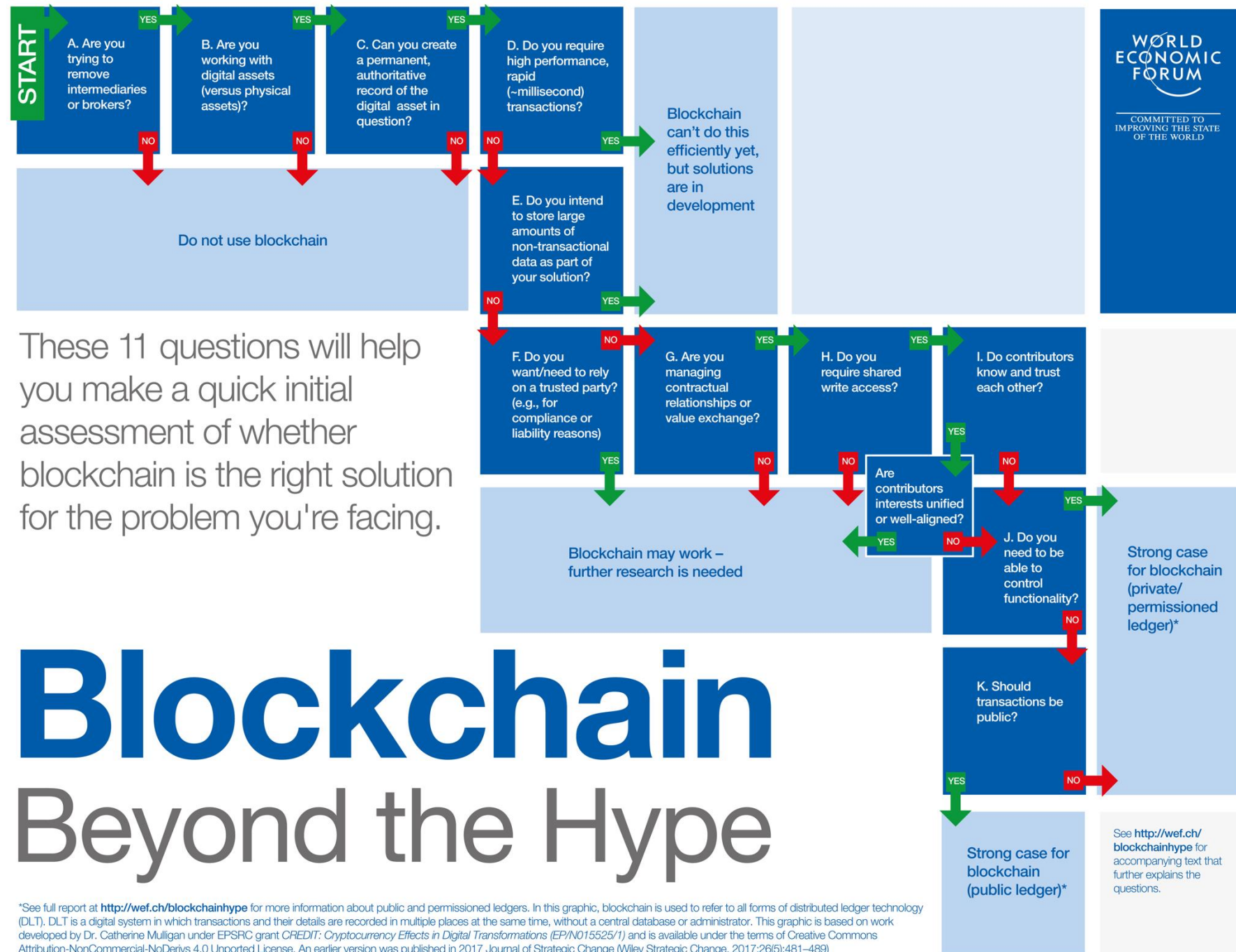


Bart Suichies model

Blockchain

WEF Model

<https://www.weforum.org/agenda/2018/04/questions-blockchain-toolkit-right-for-business>

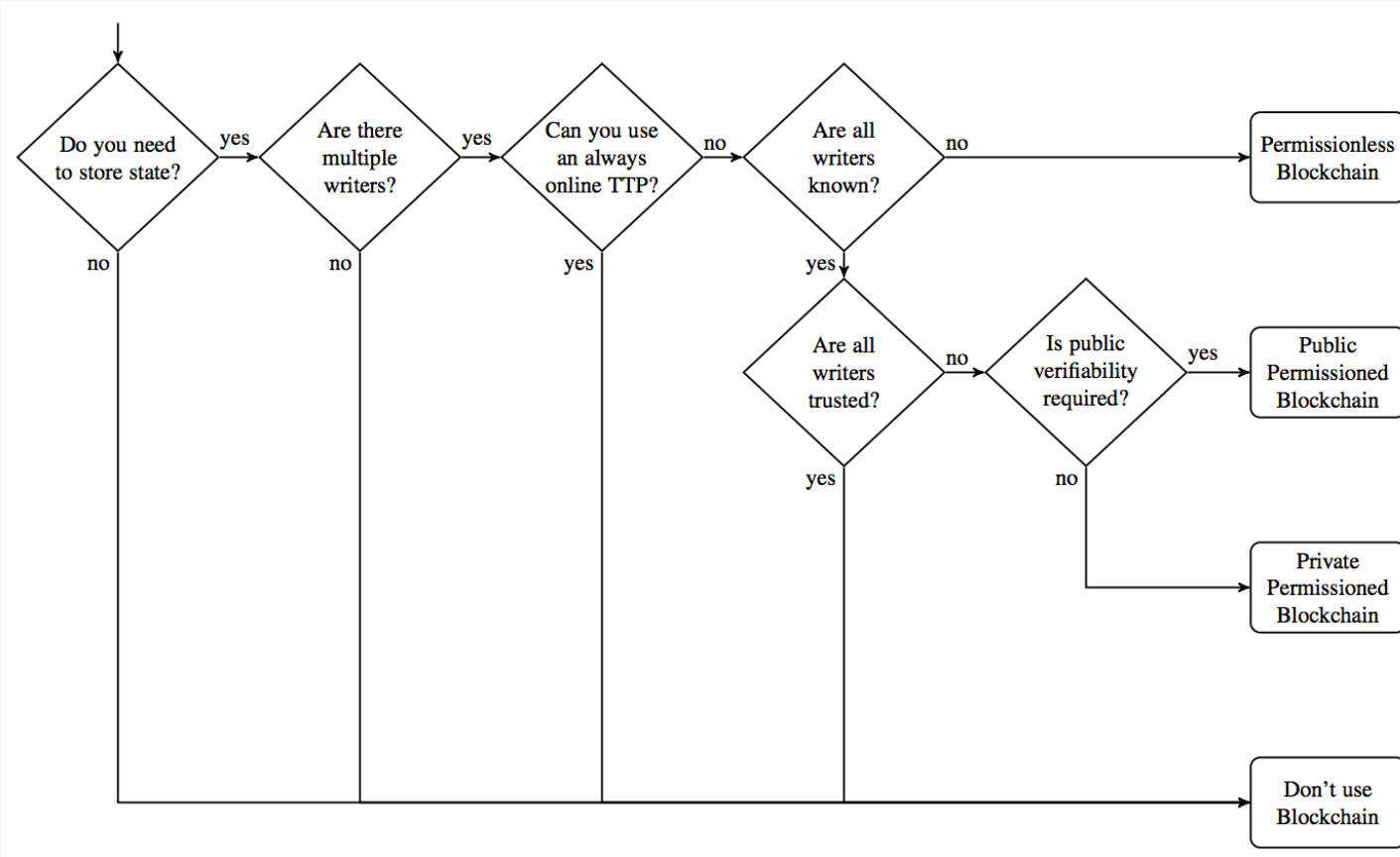


Blockchain Beyond the Hype

*See full report at <http://wef.ch/blockchainhype> for more information about public and permissioned ledgers. In this graphic, blockchain is used to refer to all forms of distributed ledger technology (DLT). DLT is a digital system in which transactions and their details are recorded in multiple places at the same time, without a central database or administrator. This graphic is based on work developed by Dr. Catherine Mulligan under EPSRC grant *CREDIT: Cryptocurrency Effects in Digital Transformations (EP/N015525/1)* and is available under the terms of Creative Commons Attribution-NonCommercial-NoDerivs 4.0 Unported License. An earlier version was published in 2017 *Journal of Strategic Change* (Wiley Strategic Change. 2017;26(5):481–489).

Blockchain feasibility

Model by Karl
Wüst & Arthur
Gervais



Question?

